

■■■■ ■■■■ ■■■■■ ■■■■■■ ■■■■■■■ ■■■■■■■■ ■■■■■■■■■ ■■■■■■■■■■ — ■■■■■■ ■■■■■■ ■■■■■■■ ■■■■■■■■ ■■■■■■■■■ ■■■■■■■■■■

Hotel & Serviced Apartments Management System — ERD & Database Design

[illegible]

Table	Columns	Columns	Columns
Table 1	39	38 + Table 2	Table 3
Table 4	VARCHAR(50)	SERIAL + UUID	Table 5
Table 6	VARCHAR(50)	SERIAL + booking_ref	Table 7
Table 8 — Table 9	JSONB (line_items)	Table 10 invoice_items Table 11	3NF Table 12
Table 13	JSONB (companions)	Table 14 booking_guests Table 15	3NF Table 16
Table 17 POS	Table 18 Table 19	4 Table 20 Table 21	Table 22 Table 23
Table 24	Table 25 Table 26	Table 27 users + role_permissions	Table 28 Table 29
Table 30	Table 31	audit_log Table 32 + trigger	Table 33 Table 34
Table 35	✓ Table 36	✓ Table 37 (Table 38)	—
iCal / Table 39 / Chatbot	✓ Table 40 (4 Table 41)	Table 42 Table 43 Table 44 Table 45	Table 46 Table 47
HR / Table 48	✓ Table 49 (3 Table 50)	Table 51 Table 52 Table 53 Table 54	Table 55 Table 56
Table 57 KPI Table 58	daily_kpis (Table 59)	Table 60 Table 61	Table 62 Table 63
Table 64 Table 65 Assets	✓ Table 66	Table 67 Table 68	Table 69 Table 70
Table 71 Table 72	✓ Table 73 (5 Table 74)	Table 75 Table 76	Table 77 Table 78

■■■■■ ■■■■■:	1:M — ■■■■ ■■ ■■■■	M:M — ■■■■ ■■ ■■■■ (■■■■ ■■■■ ■■■■)	1:1 — ■■■■ ■■ ■■■■
■■■■ ■■■■■:	PK = ■■■■ ■■■■ (■■■■)	FK = ■■■■ ■■■■ (■■■■■)	★ NEW = ■■■■ ■■■■ ■■ ■■■■■ ■■■■■

[illegible]

clients		
PK	id	SERIAL
	public_id	UUID
	name	VARCHAR(200)
	facility_type	VARCHAR(50)
	email	VARCHAR(255)
	phone	VARCHAR(20)
	status	VARCHAR(20)
FK	plan_id	INTEGER
	enabled_modules	JSONB
	created_at	TIMESTAMPZ

subscription_plans	
PK id	SERIAL
name	VARCHAR(100)
price_monthly	NUMERIC(10,2)
max_rooms	INTEGER
max_users	INTEGER
allowed_modules	JSONB
is_active	BOOLEAN

subscriptions	
PK id	SERIAL
FK client_id	INTEGER
FK plan_id	INTEGER
status	VARCHAR(20)
start_date	DATE
end_date	DATE
amount_paid	NUMERIC(10,2)

users		★ NEW
PK	id	SERIAL
FK	client_id	INTEGER
	email	VARCHAR(255)
	full_name	VARCHAR(200)
	password_hash	VARCHAR(255)
	role	VARCHAR(50)
	is_active	BOOLEAN
	last_login_at	TIMESTAMPtz
FK	created_by	INTEGER

role_permissions		★ NEW
PK	id	SERIAL
	role	VARCHAR(50)
	resource	VARCHAR(100)
	can_create	BOOLEAN
	can_read	BOOLEAN
	can_update	BOOLEAN
	can_delete	BOOLEAN
	can_approve	BOOLEAN

branches		★ NEW
PK	id	SERIAL
FK	client_id	INTEGER
name		VARCHAR(150)
address		TEXT
is_active		BOOLEAN

[illegible]

room_types	
PK id	SERIAL
FK client_id	INTEGER
code	VARCHAR(20)
name_ar	VARCHAR(100)
base_price_night	NUMERIC(10,2)
base_price_month	NUMERIC(10,2)
max_occupancy	SMALLINT

rooms		
PK	id	SERIAL
FK	client_id	INTEGER
	room_number	VARCHAR(20)
FK	room_type_id	INTEGER
	status	VARCHAR(30)
FK	current_booking_id	INTEGER
	current_guest	VARCHAR(200)
	checkout_due	DATE
	last_action_by	VARCHAR(200)
	last_action_at	TIMESTAMPZ

amenities		★ NEW
PK	id	SERIAL
FK	client_id	INTEGER
	name_ar	VARCHAR(100)
	icon	VARCHAR(50)

room_amenities		★ NEW M:M 
PK	room_id	INTEGER
PK	amenity_id	INTEGER

PK	id	SERIAL
FK	client_id	INTEGER
	full_name	VARCHAR(200)
	id_type	VARCHAR(30)
	id_number	VARCHAR(50)
	phone	VARCHAR(20)
	vip_level	VARCHAR(20)
	blacklisted	BOOLEAN
	total_stays	INTEGER

guest_documents		★ NEW
PK	id	SERIAL
FK	guest_id	INTEGER
FK	client_id	INTEGER
	doc_type	VARCHAR(30)
	file_url	TEXT
FK	uploaded_by	INTEGER

bookings	
PK id	SERIAL
FK client_id	INTEGER
booking_ref	VARCHAR(30)
FK room_id	INTEGER
FK primary_guest_id	INTEGER
check_in_date	DATE
check_out_date	DATE
actual_check_in	TIMESTAMPZ
actual_check_out	TIMESTAMPZ
booking_type	VARCHAR(20)
status	VARCHAR(30)
rate_per_night	NUMERIC(10,2)
total_nights	SMALLINT
net_amount	NUMERIC(12,2)
FK created_by	INTEGER

booking_guests		★ NEW
		M:M
PK	booking_id	INTEGER
PK	guest_id	INTEGER
	is_primary	BOOLEAN
FK	added_by	INTEGER

booking_services		★ NEW
PK	id	SERIAL
FK	booking_id	INTEGER
	service_name	VARCHAR(200)
	quantity	NUMERIC(8,2)
	unit_price	NUMERIC(10,2)
	total_price	NUMERIC(12,2)

room_actions	
PK id	SERIAL
FK client_id	INTEGER
FK room_id	INTEGER
action_type	VARCHAR(50)
old_status	VARCHAR(30)
new_status	VARCHAR(30)
staff_name	VARCHAR(200)
created_at	TIMESTAMPTZ

room_types 1 ■■■■ * rooms	rooms * ■■■■ * amenities (via room_amenities)	rooms 1 ■■■■ * bookings	guests * ■■■■ * bookings (via booking_guests)
bookings 1 ■■■■ * booking_services	rooms 1 ■■■■ * room_actions	guests 1 ■■■■ * guest_documents	rooms.room_type_id ■■FK■■→ room_types

invoices	
PK	id SERIAL
FK	client_id INTEGER
	invoice_number VARCHAR(30)
FK	booking_id INTEGER
FK	guest_id INTEGER
	invoice_date DATE
	subtotal NUMERIC(12,2)
	discount_amount NUMERIC(10,2)
	tax_rate NUMERIC(5,2)
	tax_amount NUMERIC(10,2)
	total_amount NUMERIC(12,2)
	amount_paid NUMERIC(12,2)
	balance_due NUMERIC(12,2)
	status VARCHAR(20)
FK	issued_by INTEGER

invoice_items		★ NEW
PK	id	SERIAL
FK	invoice_id	INTEGER
	description	VARCHAR(500)
	item_type	VARCHAR(30)
	quantity	NUMERIC(8,2)
	unit_price	NUMERIC(10,2)
	discount_pct	NUMERIC(5,2)
	total_price	NUMERIC(12,2)

payments		■■■■■■■■■■
PK	id	SERIAL
FK	client_id	INTEGER
FK	invoice_id	INTEGER
	amount	NUMERIC(12,2)
	payment_method	VARCHAR(50)
	payment_date	TIMESTAMPZ
	reference_number	VARCHAR(100)
FK	received_by	INTEGER

pos_categories		★ NEW POS
PK	id	SERIAL
FK	client_id	INTEGER
	name	VARCHAR(100)
	sort_order	SMALLINT
	is_active	BOOLEAN

pos_items		★ NEW POS
PK	id	SERIAL
FK	client_id	INTEGER
FK	category_id	INTEGER
	name	VARCHAR(200)
	price	NUMERIC(10,2)
	tax_rate	NUMERIC(5,2)
	deduct_inventory	BOOLEAN
FK	inventory_item_id	INTEGER

pos_orders		
PK	id	SERIAL
FK	client_id	INTEGER
	order_number	VARCHAR(30)
FK	booking_id	INTEGER
FK	guest_id	INTEGER
	order_type	VARCHAR(30)
	status	VARCHAR(20)
	total	NUMERIC(12,2)
	bill_to_room	BOOLEAN
FK	cashier_id	INTEGER

pos_order_items		★ NEW POS
PK	id	SERIAL
FK	order_id	INTEGER
FK	item_id	INTEGER
	item_name	VARCHAR(200)
	quantity	NUMERIC(8,2)
	unit_price	NUMERIC(10,2)
	total_price	NUMERIC(12,2)

■■■■■■■■■ — ■■■■■■■■ ■POS			
bookings 1 ■■■■■ * invoices	invoices 1 ■■■■■ * invoice_items	invoices 1 ■■■■■ * payments	pos_categories 1 ■■■■■ * pos_items
pos_items 1 ■■■■■ * pos_order_items	pos_orders 1 ■■■■■ * pos_order_items	bookings 1 ■■■■■ * pos_orders	pos_items.inv_item_id ■■■FK■■■→ inventory_items

housekeeping_task

PK	id	SERIAL
FK	client_id	INTEGER
FK	room_id	INTEGER
FK	booking_id	INTEGER
	task_type	VARCHAR(50)
	status	VARCHAR(20)
	priority	VARCHAR(20)
FK	assigned_to	INTEGER
	started_at	TIMESTAMPZ
	completed_at	TIMESTAMPZ

maintenance_requests		■■■■■■■■
PK	id	SERIAL
FK	client_id	INTEGER
FK	room_id	INTEGER
	issue_type	VARCHAR(100)
	description	TEXT
	priority	VARCHAR(20)
	status	VARCHAR(20)
FK	assigned_to	INTEGER
	resolved_at	TIMESTAMPZ

inventory_items	
PK	id SERIAL
FK	client_id INTEGER
FK	category_id INTEGER
	name VARCHAR(200)
	sku VARCHAR(50)
	unit VARCHAR(30)
	current_stock NUMERIC(12,2)
	min_stock NUMERIC(12,2)
	unit_cost NUMERIC(10,2)

inventory_transactions

PK	id	SERIAL
FK	client_id	INTEGER
FK	item_id	INTEGER
	transaction_type	VARCHAR(20)
	quantity	NUMERIC(12,2)
	unit_cost	NUMERIC(10,2)
	reference_type	VARCHAR(30)
FK	performed_by	INTEGER
	transaction_date	TIMESTAMPZ

suppliers	
PK id	SERIAL
FK client_id	INTEGER
name	VARCHAR(200)
vat_number	VARCHAR(50)
contact_phone	VARCHAR(20)
payment_terms	INTEGER
is_active	BOOLEAN

purchase_orders	
PK	id SERIAL
FK	client_id INTEGER
	po_number VARCHAR(30)
FK	supplier_id INTEGER
	status VARCHAR(20)
	total_amount NUMERIC(12,2)
FK	created_by INTEGER
FK	approved_by INTEGER

marketers	
PK id	SERIAL
name	VARCHAR(200)
email	VARCHAR(255)
ref_code	VARCHAR(20)
commission_rate	NUMERIC(5,2)
status	VARCHAR(20)
total_referrals	INTEGER

marketer_referrals	
PK id	SERIAL
FK marketer_id	INTEGER
FK client_id	INTEGER
ref_code_used	VARCHAR(20)
commission_earned	NUMERIC(10,2)
commission_paid	BOOLEAN

audit_log		NEW
PK	id	BIGSERIAL
FK	client_id	INTEGER
FK	user_id	INTEGER
	user_email	VARCHAR(255)
	action	VARCHAR(100)
	resource_type	VARCHAR(100)
	resource_id	TEXT
	old_values	JSONB
	new_values	JSONB
	ip_address	INET
	created_at	TIMESTAMPTZ

inventory_categories	
id	SERIAL
client_id	INTEGER
name	VARCHAR(100)

■■■■■■■ — ■■■■■■ ■■■■■■			
rooms 1 ■■■■ * housekeeping_tasks	rooms 1 ■■■■ * maintenance_requests	inventory_categories 1 ■■■■ * inventory_items	inventory_items 1 ■■■■ * inventory_transactions
suppliers 1 ■■■■ * purchase_orders	marketers 1 ■■■■ * marketer_referrals	clients 1 ■■■■ 1 marketer_referrals	audit_log: append-only (trigger)

